

Dancing on the Saddles: A Geometric Framework for Stochastic Equilibrium Dynamics[†]

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Abstract

Extending saddle-path analysis to stochastic environments, I introduce *conditional saddle paths*: equilibrium trajectories under frozen exogenous regimes, traced by stochastic-equilibrium decision rules. I first use their *geometry* to diagnose state dependence: shock responses depend on position along the saddle and on slope gaps across regimes. I then show that their *monotonicity* delivers dimension reduction, since aggregate capital exactly indexes the equilibrium state within a frozen-regime spell in [Krusell and Smith \(1998\)](#). Across spells, a synchronization theorem bounds the residual gap between histories by the regimes they recently shared. Finally, extending the geometry to a two-dimensional conditional saddle *surface* with an investment externality, I analyze the optimal investment tax credit surface over the state space.

Keywords: Conditional saddle path, business cycles, state-dependent dynamics, sufficient statistics, heterogeneous agents, investment externality.

JEL codes: C62, D58, E32.

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1 Introduction

Understanding equilibrium dynamics in models with aggregate uncertainty remains a central challenge in macroeconomics. Unlike deterministic models, where saddle-path diagrams provide immediate geometric intuition, stochastic equilibrium models lack comparable geometric frameworks. This makes it difficult to develop intuition about how economies respond to shocks, how different states interact, and why certain computational methods work. The challenge becomes particularly acute in heterogeneous-agent models, where the natural state variable is an infinite-dimensional wealth distribution, yet computational work routinely achieves dimension reduction to low-dimensional aggregates. Can we extend geometric saddle-path analysis to stochastic environments? What does such a framework reveal about equilibrium dynamics, state-dependent responses, and the success of computational approximations?

The paper makes three contributions, all drawn from a single geometric object. First, I define *conditional saddle paths*: equilibrium trajectories under frozen exogenous regimes, traced by the decision rules of the stochastic equilibrium. Each regime's trajectory draws a policy graph in the aggregate phase plane, decomposing fluctuations into movements *along* a conditional saddle (endogenous propagation) and *across* saddles (exogenous shocks), the stochastic analogue of the deterministic phase diagram. The geometry maps to the economics: shock impacts are state-independent when the saddles are parallel; relative tilt makes the same shock hit harder in some states; and even with parallel saddles, the response *shape* (peak timing and persistence) depends on position along the saddle. The graphs also visualize equilibrium-level comparative statics: a changed parameter (e.g., risk aversion) re-solves the recursive competitive equilibrium and tilts the saddles, providing economic insights on its role over the whole equilibrium dynamics. I develop these in a canonical RBC model (Sections 2–3).

Second, the same geometry speaks to a long-standing computational puzzle: why tracking a single aggregate can stand in for an entire wealth distribution. When an aggregate variable moves monotonically along the conditional saddle, it exactly indexes the equilibrium state within a frozen-regime spell, so one number recovers the distribution and every price. I show that aggregate capital plays this role in [Krusell and Smith \(1998\)](#), near the conditional steady state and, under checkable conditions, on an explicit region away from it. A cheap order-preservation check on the computed frozen-regime map, satisfied throughout the [Krusell and Smith \(1998\)](#)

economy, certifies the key condition (Section 4.1). Exactness is within-spell only. Two histories can reach the same K carrying different distributions, and how close those distributions are depends on how fast the economy forgets its regime history.

Third, I extend the geometry to two predetermined states, where the conditional saddle becomes a *surface*. In an RBC economy where investment builds knowledge that firms do not pay for (Section 4.2), the planner traces a second surface, resting at higher capital and knowledge, and a state-contingent investment tax credit implements it. That credit is itself a surface over the two stocks, so optimal policy can be analyzed geometrically on the same state space. At the illustrative spillover it is 2.4 percent of gross investment, nearly proportional to the spillover elasticity, and steady-state matching remains distinct from transition implementation.

Related literature This paper contributes to three strands of literature in macroeconomics. The first studies global equilibrium dynamics and solution methods under aggregate uncertainty, where characterizing equilibrium dynamics with aggregate shocks has motivated extensive methodological development. Marcet (1988), Den Haan and Marcet (1990), and Krusell and Smith (1998) pioneered bounded rationality approximations; Krusell and Smith (1998) found that a simple linear forecasting rule in aggregate capital achieves remarkable accuracy ($R^2 > 0.9999$) despite the infinite-dimensional state space. Subsequent work sharpened accuracy and efficiency through perturbation and projection methods (Reiter, 2010; Ahn et al., 2018; Levintal, 2018; Winberry, 2018; Bayer and Luetticke, 2020; Bhandari et al., 2023), simulation and aggregation approaches (Den Haan, 1996; Den Haan and Rendahl, 2010; Judd et al., 2011; Maliar and Maliar, 2015), adaptive sparse grids (Brumm and Scheidegger, 2017), sequence-space methods (Auclert et al., 2021; Azinovic-Yang and Žemlička, 2025), and deep learning (Maliar et al., 2021; Azinovic et al., 2022; Fernández-Villaverde et al., 2023; Han et al., 2026). Rather than reducing the state, Childers (2018) linearizes in function space, treating the entire distribution as the state, with consistency guarantees. On the theoretical side, Miao (2006) and Cao (2020) establish existence of competitive and recursive equilibria in this class; Proehl (2025) gives conditions for existence and uniqueness under aggregate and idiosyncratic risk, and Proehl (2024) develops a companion global solution method. The framework here is complementary, taking the recursive equilibrium as given and characterizing the geometry of its dynamics.

The conditional steady state links this framework to Markov-switching DSGE. Bianchi (2016) defines it as the value a state approaches while a regime persists, and Chung et al. (2007) study the same object as the regime-dependent steady state, where it anchors regime-dependent decision rules.¹ Here the point is instead the limit of an orbit: the conditional saddle path is the closure of the frozen-regime trajectory, so the steady state is where the path ends rather than where the analysis begins. Proposition 4 orders it against its perfect-foresight counterpart, and monotonicity along the path turns it into a coordinate.

The sequence-space Jacobian method of Auclert et al. (2021) reduces dimension by linearizing around a steady state; conditional saddle paths instead operate globally, exploiting monotonicity of an aggregate coordinate along the frozen-regime path. The two coincide to first order near a conditional steady state, but away from it the saddle bends, capturing state-dependent dynamics that first-order methods cannot, such as the regime-dependent slope asymmetries frictions generate (Section 3).

This paper differs from these computational contributions by providing geometric foundations for *why* dimension reduction works: aggregate capital’s within-spell sufficiency follows from nullcline invariance and monotonicity rather than numerical coincidence. This also has computational implications, since along each conditional saddle aggregate capital provides a theoretically justified metric for the repeated transition method of Lee (2026a).²

Second, this paper extends the geometric tradition of phase-diagram analyses in growth theory (Solow, 1956; Swan, 1956; Ramsey, 1928; Koopmans, 1963; Cass, 1965; Barro and Sala-i-Martin, 1992) to stochastic environments. Relatedly, the stable transformation path of Buera et al. (2024) extends the balanced growth path, and its turnpike property, to non-stationary growth models, defining a stable path independent of initial conditions. Conditional saddle paths extend the same classical geometry along a different dimension, from deterministic to stochastic settings, providing a vi-

¹Foerster et al. (2016) show that no single point is stationary under every regime and across every transition. Each K_A^{cs} is stationary under its own regime alone, and its failure to be stationary after a switch is what carries the economy across saddles.

²Freezing the aggregate state and analyzing transitions appears in several computational approaches to heterogeneous-agent models, including Bourany (2018), Section 4.3 of a working paper version of Achdou et al. (2021), and Lee (2026a). In continuous time, the master-equation literature (Achdou et al., 2021; Bilal, 2023) treats the distribution itself as the state and asks when finite-dimensional projections suffice, the rigorous mathematical home of this question. Conditional saddle paths formalize the underlying geometric objects, and the sufficiency results explain when dimension reduction is exact along frozen-regime paths.

sual tool for how microfounded frictions generate state-dependent dynamics (Kaplan and Violante, 2014; Vavra, 2014; Berger and Vavra, 2015; Basu and Bundick, 2017; Bloom et al., 2018; Kaplan et al., 2018; Petrosky-Nadeau et al., 2018; Rognlie et al., 2018; Baley and Blanco, 2019; Pizzinelli et al., 2020; Berger et al., 2021; Winberry, 2021; Lee, 2026b; Melcangi, 2024).

Finally, the random dynamical systems literature (Arnold, 1998; Schenk-Hoppé, 1998, 2001) focuses on history-dependent stochastic saddle paths (Yannacopoulos, 2011) that vary with the realized shock sequence.³ By contrast, conditional saddle paths are history-invariant geometric objects, the direct analogue of deterministic phase diagrams.

2 Conditional saddle path

2.1 Definitions and assumptions

I consider a generic dynamic stochastic model whose recursive competitive equilibrium (RCE) is characterized by an aggregate state S and laws of motion $(\Gamma_{\text{endo}}, \Gamma_{\text{exo}})$:

$$S = [\Phi, A] \tag{1}$$

where Φ is the endogenous and A the exogenous aggregate state, the latter possibly multivariate. The exogenous law Γ_{exo} is a Markov chain; for the illustration I take a two-state chain with $A \in \{B, G\}$ and $\Gamma_{\text{exo}}(A'|A) > 0 \forall A', A$.

The distributional state space $\mathcal{X}$ has elements Φ summarizing the cross-sectional distribution of payoff-relevant idiosyncratic household states (e.g., assets and productivity types) plus any endogenous objects needed to evaluate decision rules. Formally, $\mathcal{X}$ is a subset of a metric space of probability measures, possibly augmented by a finite-dimensional vector of aggregate variables. Its metric d treats two states as close when the averages they assign to any well-behaved function of the individual state are close, with the finite-dimensional components compared in the Euclidean norm.⁴ Throughout, $\Gamma_{\text{endo}}(\cdot, A) : \mathcal{X} \rightarrow \mathcal{X}$ maps the current distributional state to

³In Random Dynamical Systems language, the conditional steady state is a deterministic realization of the random fixed point (Schenk-Hoppé and Schmalfuß, 2001), and the conditional saddle path is the invariant manifold of the frozen-regime dynamics. The synchronization theorem of Section 4.1 (Theorem 2) uses exactly this pullback random-fixed-point construction, recovering the current state as the long-history limit of trajectories started ever further in the past under the same regime history.

⁴ d is any metric for which $d(\Phi_n, \Phi) \rightarrow 0$ if and only if $\int f d\Phi_n \rightarrow \int f d\Phi$ for every bounded

next period's under frozen regime A .

The central object formalizes a natural thought experiment: the economy sits at (Φ_0, A) today and the regime stays at A indefinitely. The resulting equilibrium trajectory, computed under decision rules that fully internalize future regime changes, is a frozen-regime orbit. Its closure (visited states plus their limit points) is the conditional saddle path. In aggregate phase diagrams the associated regime-specific locus is a conditional policy graph, to which the slope and curvature statements below refer.

Definition 1 (Frozen-regime orbit and conditional saddle path).

Fix a regime $A \in \{B, G\}$ and an initial state $\Phi_0 \in \mathcal{X}$. Let $\{\Phi_t\}_{t \geq 0}$ denote the frozen-regime continuation under A , defined recursively by

$$\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A), \quad t \geq 0. \quad (2)$$

The frozen-regime orbit is

$$\mathcal{O}(\Phi_0, A) := \{\Phi_t : t \geq 0\}. \quad (3)$$

The conditional saddle path under A from Φ_0 is defined as the closure of the frozen-regime orbit:

$$\mathcal{M}(\Phi_0, A) := \overline{\mathcal{O}(\Phi_0, A)}. \quad (4)$$

Two projection components recur: a *coordinate* $e : \mathcal{X} \rightarrow \mathbb{R}$, a predetermined aggregate variable such as capital K or a knowledge stock H ; and an *aggregate object* $g(\Phi, A)$, an equilibrium outcome such as consumption or investment.

Definition 2 (Conditional policy graph).

For a regime A , a coordinate e , and an aggregate object g , the conditional policy graph in the (e, g) projection is

$$\mathcal{P}_A^g := \{(e(\Phi), g(\Phi, A)) : \Phi \in \mathcal{D}_A\}, \quad (5)$$

where $\mathcal{D}_A$ is the set of distributional states reached by the frozen- A continuations under study.

When e is injective (one-to-one) on a given conditional saddle path, the restriction of $\mathcal{P}_A^g$ to that path is the graph of a single-valued function of e .

Based on the conditional saddle path, I define the conditional steady state:

continuous function f ; this is the standard notion of weak convergence.

Definition 3 (Conditional steady state).

Given (Φ_0, A) , if the frozen-regime continuation $\{\Phi_t\}_{t \geq 0}$ converges, I denote its limit by

$$\Phi^{cs}(\Phi_0, A) := \lim_{t \rightarrow \infty} \Phi_t. \quad (6)$$

Fixing A makes the law of motion deterministic on the distribution space: from Φ_0 , $\{\Phi_t\}_{t \geq 0}$ is generated by repeated application of $\Gamma_{\text{endo}}(\cdot, A)$. Thus $\mathcal{M}(\Phi_0, A)$ is the natural analogue of the saddle arm in deterministic saddle-path analysis, isolating the equilibrium states (and their limit points) visited along the convergence dynamics under regime A .⁵

For the conditional saddle to serve as a useful geometric object, a one-dimensional curve traversed during each frozen-regime spell, the dynamics must converge to a unique limit and vary smoothly. The following assumption collects these conditions.

Assumption 1 (Regularity of conditional saddle paths).

Fix a regime $A \in \{B, G\}$ and an initial state $\Phi_0 \in \mathcal{X}$. Let $\{\Phi_t\}_{t \geq 0}$ be defined by $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$.

- (i) (Unique existence) The conditional steady state $\Phi^{cs}(\Phi_0, A)$ uniquely exists on $\mathcal{M}(\Phi_0, A)$.
- (ii) (Continuity) The map $\Gamma_{\text{endo}}(\cdot, A) : \mathcal{X} \rightarrow \mathcal{X}$ is continuous. Moreover, the aggregate variables (e.g., aggregate capital K and consumption C) vary continuously in Φ when restricted to $\mathcal{M}(\Phi_0, A)$.

Part (i) requires that, conditional on regime A and initial state Φ_0 , the frozen-regime continuation converge to a unique conditional steady state; it asserts neither global uniqueness across initial conditions nor the absence of multiplicity across regimes.⁶ Part (ii) imposes continuity of the law of motion and aggregate observables on the conditional saddle, so small perturbations of the state produce small changes in aggregate outcomes. Assumption 1(i) thus restricts attention to convergent continuations, ruling out periodic orbits or divergent paths. In the RBC benchmark

⁵Strictly, in discrete time $\mathcal{M}(\Phi_0, A) = \overline{\mathcal{O}(\Phi_0, A)}$ is an orbit closure rather than a smooth manifold, and the saddle-path label refers to its path-like role. Slope and curvature therefore refer to the policy graph $\mathcal{P}_A^g$. In continuous time the frozen-regime object is a one-dimensional invariant manifold.

⁶Proehl (2025) establishes global existence and uniqueness of the recursive equilibrium under explicit conditions on primitives, whereas Assumption 1(i) allows global multiplicity as in Walsh and Young (2024); conditional on a given initial state and fixed regime, the equilibrium path is assumed uniquely determined.

convergence follows from diminishing returns: the marginal product of capital is high below the conditional steady state and low above it, so capital approaches it from either side. In the heterogeneous-household economy convergence is instead a maintained condition on the equilibrium law of motion.

With this regularity the conditional saddle admits a natural notion of position along the orbit. Theorem 1 (Section 4.1) formalizes the key implication: if some aggregate variable moves strictly one-way along the path, it provides a valid coordinate for the entire conditional saddle, yielding an exact one-dimensional representation on $\mathcal{M}(\Phi_0, A)$. This result is *pathwise*: it holds along each orbit $\mathcal{M}(\Phi_0, A)$ separately, regardless of the number of predetermined states. What changes with two or more stocks is that the orbits over initial conditions fill a *surface* rather than a curve, on which a single monotone coordinate need not exist. This is the multi-state case Section 4.2 analyzes, where the surface also holds the paper’s planner–equilibrium geometry.

2.2 Representative-agent economy: A canonical RBC

A representative household with log utility maximizes lifetime utility under stochastic aggregate TFP A_t , given (a_0, A_0) . I use the recursive competitive equilibrium, equivalent to the sequential equilibrium under standard regularity conditions. The recursive form is:

$$v(a; K, A) = \max_{c, a'} \log(c) + \beta \mathbb{E}v(a'; K', A') \quad (7)$$

$$c + a' = a(1 + r(K, A)) + w(K, A) \quad (8)$$

$$a' \geq -\bar{a} \quad (9)$$

$$K' = \Gamma_{\text{endo}}(K, A), \quad A' \sim \Gamma_{\text{exo}}(A'|A), \quad (10)$$

where v is the value function, a beginning-of-period wealth, K aggregate capital, A aggregate TFP, and Γ_{endo} the law of motion for K . TFP A follows a Markov-switching process between levels B and G , with $G > B$:

$$\Gamma_{\text{exo}} = \begin{bmatrix} \pi_{BB} & \pi_{BG} \\ \pi_{GB} & \pi_{GG} \end{bmatrix} \quad (11)$$

Competitive factor prices from a CRS Cobb–Douglas production function are:

$$r(K, A) = A\alpha K^{\alpha-1} - \delta \quad (12)$$

$$w(K, A) = A(1 - \alpha)K^\alpha, \quad (13)$$

where capital K satisfies $K = a$ in equilibrium. The recursive competitive equilibrium (RCE) is:

Definition 4 (Recursive competitive equilibrium).

$(c, a', v, r, w, \Gamma_{\text{endo}})$ is a recursive competitive equilibrium if:

1. given prices (r, w) , the policies (c, a') attain the value v in the household problem;
2. prices satisfy the competitive factor-price conditions above;
3. consistency holds:

$$a'(K; K, A) = K' = \Gamma_{\text{endo}}(K, A). \quad (14)$$

Using Γ_{endo} from Definition 4, I define the conditional K -nullcline.

Definition 5 (Conditional K -nullcline in RBC).

Fix a regime $A \in \{B, G\}$. The conditional K -nullcline is the zero-net-investment locus in the (K, C) phase space,

$$\mathcal{N}(A) := \{(K, C) \in \mathbb{R}_{++} \times \mathbb{R}_{++} : \Delta K(K, C; A) = 0\}, \quad (15)$$

where $\Delta K(K, C; A)$ is the one-period change in capital implied by the resource constraint at (K, C) under regime A . When $\mathcal{N}(A)$ is the graph of a function, I write $\mathcal{N}(A) = \{(K, C) : C = C_A^{Knull}(K)\}$.

Along a strictly monotone frozen-regime continuation, the orbit meets $\mathcal{N}(A)$ only in the limit, at the conditional steady state; the locus is thus distinct from the *realized nullcline crossings* $\{(K_t, C_t) : t \geq 0, K_{t+1} = K_t\}$, empty before the limit. The resource constraint $Y = C + I$ with $I = K_{t+1} - (1 - \delta)K_t$ gives $\Delta K = AK^\alpha - \delta K - C$, so $\mathcal{N}(A)$ is an explicit function graph.

Proposition 1 (Characterizing the conditional K -nullcline in RBC).

The conditional nullclines of aggregate capital K for $A \in \{B, G\}$ are

$$C_A^{Knull}(K) = AK^\alpha - \delta K, \quad (16)$$

independent of any initial condition.

Proof. From $\Delta K(K, C; A) = 0$ and $\Delta K = AK^\alpha - \delta K - C$, the nullcline is $C = AK^\alpha - \delta K$ for each A . $\blacksquare$

The nullcline invokes only $\Delta K = AK^\alpha - \delta K - C$, so it bears no imprint of preferences, beliefs, or the distribution; the same holds for the induced *direction field*, whose net-investment sign $\Delta K = C_A^{Knull}(K) - C$ is fixed by accounting alone. Beliefs and heterogeneity relocate the equilibrium orbit within a phase portrait whose boundary and sign structure they cannot move. This is the common root of belief-invariance under aggregate uncertainty (Proposition 3), underlying the monotonicity diagnostic of Appendix G,⁷ and distribution-invariance in the heterogeneous-household model (Proposition 7), underlying within-spell dimension reduction.

In discrete time capital can jump a finite amount, so the projected (K, C) trajectory can develop folds or loops, pathologies absent in continuous time. I rule them out with a geometric restriction.

Assumption 2 (No segment crossing of the conditional K -nullcline).

For each $A \in \{B, G\}$, along the frozen- A continuation: for every pre-limit transition (every $t \geq 0$ with $K_{t+1} \neq K_t$), the open straight line segment in (K, C) connecting (K_t, C_t) and (K_{t+1}, C_{t+1}) does not intersect the active regime's conditional K -nullcline $\mathcal{N}(A)$:

$$((K_t, C_t), (K_{t+1}, C_{t+1})) \cap \mathcal{N}(A) = \emptyset, \quad \forall t \geq 0 \text{ with } K_{t+1} \neq K_t. \quad (17)$$

Endpoints are not required to lie off the nullcline; in particular, the conditional steady state is allowed to lie on $\mathcal{N}(A)$, and the assumption imposes no condition at the degenerate terminal transition $K_{t+1} = K_t$.

Assumption 2 ensures consecutive points cannot strictly switch sides of the K -nullcline. Its companion is *unique K -stationarity*: $K_{t+1} = K_t$ only at $\Phi_t = \Phi^{cs}(\Phi_0, A)$, which in the representative-agent model follows from Assumption 1(i) (K is Markov under the frozen regime) and is otherwise maintained as a pathwise hypothesis. Economically, Assumption 2 rules out single-period overshooting so severe that capital jumps past its stationary level and reverses, the discrete-time counterpart of the orbit meeting $\mathcal{N}(A)$ only in the limit.⁸

⁷All lettered appendices are collected in the Online Appendix.

⁸For related geometric restrictions in dynamic optimization, see Reddy et al. (2020) and, on threshold (Skiba) phenomena, Skiba (1978), Dechert and Nishimura (1983), and Wagener (2003).

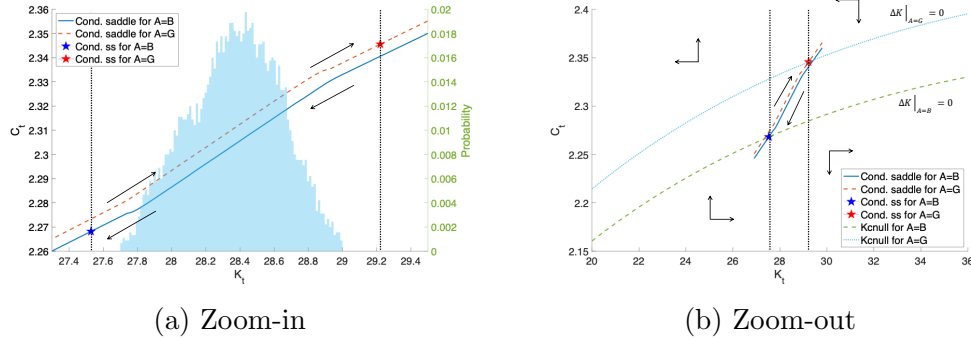
Although stated for the RBC benchmark, Assumption 2 is central to the heterogeneous-household economy (Section 4.1). It can be verified from primitives without solving the full stochastic equilibrium, conditional on a simulated frozen-regime continuation (or certified envelope bounds along it). Appendices G and E develop two complementary strategies.

Proposition 2 (*K-monotonicity in RBC*).

Fix $A \in \{B, G\}$. Under Assumptions 1 and 2, together with unique K -stationarity ($K_{t+1} = K_t$ only at $\Phi_t = \Phi^{cs}(\Phi_0, A)$ along the frozen- A continuation), aggregate capital converges to K_A^{cs} strictly monotonically.

Proof. Fix A and let $D_t := C_t - C_A^{Knull}(K_t)$ denote the nullcline gap. The resource constraint gives $\Delta K_t = -D_t$. By Assumption 2, no transition strictly switches sides of the nullcline: $D_t D_{t+1} \geq 0$ for every transition t with $K_{t+1} \neq K_t$, and $D_t \neq 0$ on any such transition. Unique K -stationarity confines $D_{t+1} = 0$ to terminal arrival at Φ_A^{cs} , so prior to convergence $\{\Delta K_t\}$ has constant nonzero sign. Unique K -stationarity rules out nonterminal flat steps ($\Delta K_t = 0$ at any $\Phi_t \neq \Phi_A^{cs}$), so every pre-convergence transition is nondegenerate and $\{K_t\}$ is strictly monotone. Convergence $K_t \rightarrow K_A^{cs}$ follows from Assumption 1. $\blacksquare$

Figure 1: Conditional saddle paths and the phase diagram



Notes: Conditional K -nullclines ($\Delta K = 0$) partition the (K, C) plane. Direction arrows show frozen-regime convergence along each conditional saddle. Panel (a) zooms into the neighborhood of the conditional steady states shown in panel (b). The histogram plots the time-series distribution of aggregate capital.

Unique K -stationarity as a no-plateau condition Unique K -stationarity rules out *plateau equilibria*: states off the conditional steady state at which aggregate capital stops moving ($\Delta K_t = 0$) while the distribution keeps evolving, excluding the

degenerate fixed points of K that arise with poverty traps, occasionally binding constraints, or policy kinks. In the representative-agent RBC, K is Markov under the frozen regime, so the condition reduces to uniqueness of the conditional steady state (Assumption 1(i)); in the heterogeneous-agent economy it holds locally via the stable eigenvalue $\lambda_A^s \in (0, 1)$ of the linearized frozen-regime dynamics (Proposition 10). A globally certified version is more delicate: the bounded price-income MPC of Assumption 3 (Appendix E) constrains the off-steady-state dynamics, but I treat unique K -stationarity as a maintained pathwise condition, checkable along any simulated frozen-regime continuation.

In the representative-agent benchmark these restrictions are not free-standing assumptions but consequences of the scalar concave structure, given two checkable inputs verified in the computed equilibrium: a curvature-margin bound and, for stochastic beliefs, a single-crossing condition on the conditional Euler residual. Since capital is the only state and next period's K depends on K alone, the frozen- A continuation iterates a single policy $g_A(K)$, and each restriction follows from a primitive:

1. strict concavity of the one-sector problem delivers a unique conditional steady state (under deterministic beliefs; the stochastic case adds the verified single-crossing input, over the capital range on which it is verified);
2. the same concavity delivers strictly monotone convergence to that steady state, with the same qualification;
3. the same concavity delivers no segment crossing of the nullcline under the curvature-margin bound: the gap between path and nullcline at consecutive dates must exceed the nullcline's bend over that step, which standard calibrations satisfy by orders of magnitude;
4. feasibility bounds capital, so explosion never arises;
5. the transversality condition enters only in selecting the stable policy.

Appendix J gives the formal proof. The conditions are stated as model assumptions because in the heterogeneous-agent economy of Section 4.1, the iterated state is the whole distribution, not scalar capital. Next period's K then depends on the distribution rather than on K alone. Monotone convergence requires a real stable

eigenvalue of the linearized distributional dynamics, and primitives do not pin one down.

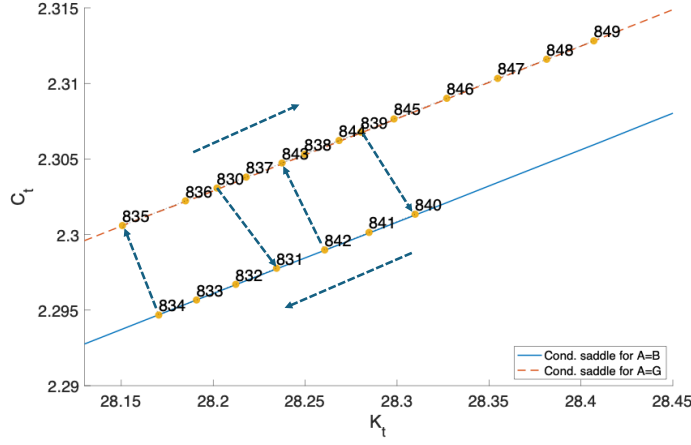
Figure 1 depicts the calibrated RBC model’s conditional saddle paths in the (K, C) plane under $A \in \{B, G\}$, solved globally via the repeated transition method of Lee (2026a).⁹ Panel (a) zooms into the conditional steady states with a histogram of simulated K_t ; panel (b) shows the global partition with conditional K -nullclines and direction arrows. All long-run dynamics stay within the region bounded by the two conditional saddles: within a spell the economy converges along the active saddle, and regime switches move it across, so the entire ergodic distribution is nested within the saddle boundaries.

MIT shock on the saddle This decomposition characterizes unexpected transitory (MIT) shocks, whose link to first-order perturbation in heterogeneous-agent models is established by Boppart et al. (2018). For shocks representable as displacements of the predetermined state under unchanged decision rules, a shift along the conditional saddle replicates the impact, and all subsequent dynamics are along-path convergence. The slope and curvature of the conditional policy graph at the displacement point thus determine propagation, including persistence, amplification, and state dependence.

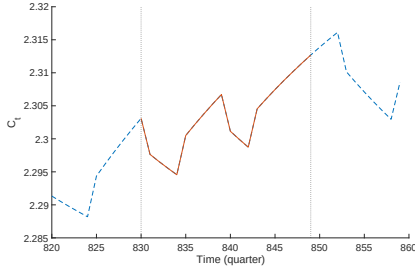
Generalized transition function Because conditional saddle paths capture all equilibrium transitions, along and across saddles, they support generalized transition function analysis (Lee, 2026a), encompassing generalized impulse response functions (Koop et al., 1996; Andreasen et al., 2017). Any shock sequence is traced in the phase diagram as along-saddle and across-saddle movements, not confined to local dynamics around a steady state.

Figure 2 shows equilibrium dynamics in the phase diagram (panel (a)) and time domain (panels (b) and (c)) over a 20-quarter subsample. Negative TFP shocks produce downward jumps in consumption across conditional saddles, followed by endogenous decline along them; positive shocks reverse this. The phase diagram captures the dynamics in a single figure and supports counterfactual analysis. For the GIRF, one can replace the two-state chain with a finely discretized AR(1) for log productivity, $\ln A' = \rho \ln A + \epsilon$, $\epsilon \sim N(0, \sigma^2)$. There is then a continuum of frozen- A

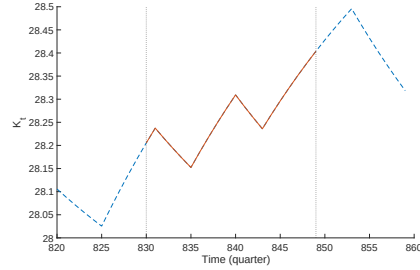
⁹For each regime A , I hold productivity fixed for 2,500 periods and iterate the equilibrium law of motion from initial capital on both sides of K_A^{cs} , recovering the full conditional saddle in the phase diagram.



(a) On the saddles



(b) C - time domain



(c) K - time domain

Figure 2: Equilibrium dynamics: saddle vs. time domain

Notes: Stochastic equilibrium dynamics on the conditional saddle paths (panel (a)) and in the time domain (panels (b) and (c)). A 20-quarter window from the simulated path (quarters 830–849).

conditional saddles indexed by A , and the GIRF path moves both along a given saddle (endogenous propagation) and across the family as A evolves.

Following a positive TFP shock, consumption jumps upward across conditional saddles on impact; thereafter, along- and across-saddle forces jointly generate a bow-shaped trajectory in the phase diagram. Capital is hump-shaped as the shock mean-reverts, and consumption first rises then falls as TFP crosses the threshold where along-saddle adjustment reverses direction.

Comparison with the perfect-foresight saddles In the *perfect-foresight* (*PF*) economy, agents believe the current regime persists forever, so the PF steady states solve $1 = \beta(1 - \delta + \alpha A(K_A^{pf})^{\alpha-1})$. Along the *conditional saddle* (*CS*), the regime is fixed ex post but agents correctly anticipate switches under Γ_{exo} , so the conditional

steady states K_A^{cs} are pinned by probability-weighted Euler equations.¹⁰ The PF and CS economies share the same K -nullcline (Proposition 3) but differ in consumption dynamics.

Proposition 3 (K -nullcline invariance over beliefs).

Conditional K -nullclines are identical between the RBC model with aggregate uncertainty and the perfect-foresight counterpart.

Proof.

From the stationary condition $\delta K = I$ and the aggregate resource constraint $Y = C + I$, equation (16) is immediate. Neither condition involves expectations or belief structures, so the nullcline is identical whether or not there is aggregate uncertainty. ■

PF–CS differences thus come entirely from consumption dynamics, not the feasibility locus; Proposition 4 formalizes this via steady-state orderings. This invariance is an accounting identity independent of beliefs and, in Section 4.1, of *who* saves, on which both the dimension-reduction result and the monotonicity diagnostic rest.

Proposition 4 (Aggregate uncertainty and the conditional steady states).

Let $R(A, K) := 1 - \delta + \alpha AK^{\alpha-1}$ denote the gross return and $c(K, A')$ the stochastic-equilibrium consumption function. Suppose that (i) at each regime's PF steady state, the consumption response to a regime switch dominates the return response (jump dominance),

$$\frac{c(K_B^{pf}, G)}{c_B^{pf}} > \frac{R(G, K_B^{pf})}{R(B, K_B^{pf})}, \quad \frac{c(K_G^{pf}, B)}{c_G^{pf}} < \frac{R(B, K_G^{pf})}{R(G, K_G^{pf})}; \quad (18)$$

and (ii) each conditional Euler residual F_A , the residual of the conditional steady-state Euler equation viewed as a function of K (Appendix A), is strictly positive below K_A^{cs} and strictly negative above it. Then

$$K_B^{cs} < K_B^{pf} < K_G^{pf} < K_G^{cs}, \quad c_B^{cs} < c_B^{pf} < c_G^{pf} < c_G^{cs}. \quad (19)$$

Proof.

See Appendix A. ■

¹⁰Specifically, $\beta\pi_{AA}(1 + \alpha A(K_A^{cs})^{\alpha-1} - \delta) + \beta\pi_{AA'}(c(K_A^{cs}, A')/c_A^{cs})^{-1}(1 + \alpha A'(K_A^{cs})^{\alpha-1} - \delta) = 1$ for each $A \in \{B, G\}$, where $A' \neq A$.

Both hypotheses are explicit and directly checkable: a persistent regime switch barely moves the gross return (only the marginal-product term shifts) while consumption moves roughly one-for-one, so condition (i) fails only in degenerate cases. Under full depreciation ($\delta = 1$) the decision rules collapse to the belief-independent form of Brock and Mirman (1972), condition (i) holds with equality, and the two sets of steady states coincide (Appendix A); away from that corner the orderings hold in the computed equilibrium.

PF steady states are nested within the CS steady states: hedging against regime switches pushes the conditional steady states apart (Deaton, 1991; Carroll, 1997). Conversely, the conditional saddle *paths* are nested within the PF paths, since at any K hedging compresses the cross-regime consumption gap. Both nestings share one mechanism: compressed consumption responses cumulate into amplified long-run capital divergence.

Conditional boundary condition The recursive formulation imposes a stochastic analogue of the transversality condition, necessary in expectation (Kamihigashi, 2003, 2005). Its conditional counterpart is $\lim_{t \rightarrow \infty} \beta^t u'(c_t) K_t = 0$ along each frozen-regime saddle. The conditional condition, holding along every history, delivers the expectation condition. The converse fails, because the frozen branch has probability zero. The conditional condition nonetheless holds in equilibrium, since the frozen continuation iterates the recursively selected policy. Appendix J establishes this from strict concavity of the one-sector problem rather than from a terminal condition.

Economies without conditional saddle paths A conditional saddle requires a predetermined endogenous state. Without one, the economy jumps each period to the allocation implied by the current exogenous state, so the frozen- A conditional “saddle” collapses to a singleton; I call such an economy *endogenously stateless*.¹¹ Models with predetermined states (capital, habits, interest-rate smoothing, or distributional states in HANK) are endogenously stateful, and conditional saddle paths exist under mild regularity conditions.

¹¹The textbook three-equation New Keynesian model (Gali, 2008) under determinacy is endogenously stateless: $(x_t, \pi_t, i_t) = \Psi s_t$ for exogenous state $s_t := (r_t^n, u_t, \nu_t)$, so freezing s_t produces no nontrivial dynamics.

3 State dependence

Responses to the same impulse vary with the business-cycle phase: unemployment dynamics are asymmetric (Neftci, 1984), fiscal multipliers are larger in recessions (Auerbach and Gorodnichenko, 2012), and monetary policy is weaker in recessions (Tenreyro and Thwaites, 2016). I analyze this state dependence through conditional saddle paths. In any stochastic dynamic model admitting such paths, an aggregate variable's response to a shock is a shift across conditional policy graphs. The benchmark is *state independence*: the impact gap between graphs is constant in the endogenous coordinate; departures show up locally as slope differences, the first-order diagnostic.

Lemma 1 (Vertical-separation benchmark).

Let I be an interval over which the regime-specific conditional policy graphs $\mathcal{P}_{A_j}^g = \{(e, g_j(e)) : e \in I\}$, $j \in \{0, 1\}$, are single-valued. Define the impact gap between the two graphs as $\Delta_g(e; A_1, A_0) := g_1(e) - g_0(e)$. Then conditions (i) and (ii) below are equivalent; if g_0, g_1 are differentiable on I , all three are equivalent:

- (i) Δ_g is constant on I ;
- (ii) $\mathcal{P}_{A_1}^g$ is a vertical translation of $\mathcal{P}_{A_0}^g$;
- (iii) the slope gap $g'_1 - g'_0$ vanishes on I .

Proof. The equivalence of (i) and (ii) is definitional. (i) implies (iii) by differentiating the constant gap. (iii) implies (i) by the mean value theorem: $\Delta'_g = g'_1 - g'_0 = 0$ on the interval I makes Δ_g constant. ■

Lemma 1 fixes the benchmark. Proposition 5 carries the following substantive claim: the slope gap reflects primitives. Exact state independence is a knife-edge: any differential tilt across conditional policy graphs generates state-dependent shock responses. The degree of state dependence depends on the friction behind the tilt, whether real (Bachmann et al., 2013; Winberry, 2021; Lee, 2026a), financial (Melcangi, 2024), or labor market (Petrosky-Nadeau et al., 2018; Pizzinelli et al., 2020). I measure the tilt locally: each regime's saddle slope at its own conditional steady state, compared across regimes as a ratio. In the log-linearized frozen-regime RBC dynamics near K_A^{cs} , let $R_A^{cs} := 1 + f'(K_A^{cs}; A) - \delta$ denote the gross return at the conditional

steady state, $\lambda_A^s \in (0, 1)$ the stable eigenvalue, and $\psi_A := dC/dK|_{\text{saddle}, A}$ the level saddle slope at K_A^{cs} .

Proposition 5 (Slope ratio as a structural diagnostic).

The level saddle slope admits the closed form

$$\psi_A = R_A^{cs} - \lambda_A^s, \quad (20)$$

and hence the cross-regime slope ratio is

$$\mathcal{R} := \frac{\psi_B}{\psi_G} = \frac{R_B^{cs} - \lambda_B^s}{R_G^{cs} - \lambda_G^s}. \quad (21)$$

Proof. Log-linearize the frozen-regime budget constraint $K_{t+1} = f(K_t; A) + (1 - \delta)K_t - C_t$ around (K_A^{cs}, C_A^{cs}) :

$$\hat{k}_{t+1} = R_A^{cs} \hat{k}_t - \frac{C_A^{cs}}{K_A^{cs}} \hat{c}_t. \quad (22)$$

On the stable subspace of the linearized dynamics, the span of the stable eigenvector, the deviations satisfy $\hat{c}_t = \psi_A^{\log} \hat{k}_t$ for some log slope $\psi_A^{\log}$. Substituting into (22) yields $\hat{k}_{t+1} = [R_A^{cs} - \frac{C_A^{cs}}{K_A^{cs}} \psi_A^{\log}] \hat{k}_t$, which by definition equals $\lambda_A^s \hat{k}_t$. Solving, $\psi_A^{\log} = (K_A^{cs}/C_A^{cs})(R_A^{cs} - \lambda_A^s)$. Converting to level units via $dC/dK|_{K^{cs}} = (C_A^{cs}/K_A^{cs}) \psi_A^{\log}$ gives (20), and (21) follows by taking the ratio.

Under perfect-foresight expectations, the steady-state Euler condition gives $R_A^{cs} = 1/\beta$ and the formula reduces to $\psi_A = 1/\beta - \lambda_A^s$. Under stochastic expectations over regime transitions, R_A^{cs} is pinned down by $1 = \beta \mathbb{E}_A[(C_A^{cs}/C(\Phi_A^{cs}, A'))^\sigma (1 + f'(K_A^{cs}; A') - \delta)]$ (the return evaluated at tomorrow's capital K_A^{cs} under the realized regime A') and generally differs from $1/\beta$. The formula remains valid with R_A^{cs} evaluated at the (stochastic) conditional steady state. ■

Proposition 5 gives $\mathcal{R}$ a structural interpretation: each regime's saddle slope is the gap between the gross return on capital at its conditional steady state and the stable eigenvalue. Both are computable from a local linearization of the frozen-regime dynamics without solving the global policy. One reads off (R_A^{cs}, λ_A^s) and applies (21). Local departures from $\mathcal{R} = 1$ indicate state-dependent impact responses. With $\mathcal{R} > 1$ the bad-regime saddle is the steeper, so the gap between the graphs narrows in K and the same shock moves consumption more from a low capital stock; $\mathcal{R} < 1$ reverses the pattern. $\mathcal{R}$ is a model-implied moment identifiable in principle from state-dependent local projections (Auerbach and Gorodnichenko, 2012; Tenreyro and Thwaites, 2016).

Since $\mathcal{M}(\Phi_0, A)$ is an orbit closure, the slope, curvature, and tangent language of Propositions 5 and 6, of Section 4.1, and of Proposition 10 refers to its local C^1 parametrization near K_A^{cs} , which exists under the regularity conditions of Proposition 10 (Appendix D).

An asymmetric adjustment cost Expanding the capital stock involves installation lags, ramp-up, and congestion, while contracting it largely means letting capital depreciate. If expanding is the more costly margin, the two regimes' policy graphs need not be parallel, and once they are not, the same shock moves the economy by different amounts depending on where it starts. The investment literature points the same way. Bachmann et al. (2013) document stronger aggregate investment responsiveness in expansions, and Winberry (2021) generates procyclical responsiveness through the extensive margin.

The benchmark of Section 2.2 accommodates this with a single change, an adjustment cost whose curvature differs on the two sides of zero net investment: $\mathcal{C}(K', K) = \frac{\tilde{\mu}}{2}((K' - K)/K)^2 K$, where $\tilde{\mu} = \mu_+$ if $K' - K \geq 0$ and $\tilde{\mu} = \mu_-$ otherwise. Its two summary parameters are the mean curvature $\bar{\mu} := (\mu_+ + \mu_-)/2$ and the asymmetry ratio $r := \mu_+/\mu_-$. Two features of this specification matter. The curvature switches at net accumulation $K' = K$ rather than at gross investment $I = K' - (1 - \delta)K$, so the friction bites when the economy stops adding to its capital stock, not when it stops investing altogether. And the cost is continuously differentiable at that threshold: only the second derivative jumps, so the asymmetry works through curvature and not through a kink in the cost or its slope. I read $\mu_+ > \mu_-$ as a reduced-form expansion bottleneck, capturing installation, ramp-up, or congestion, rather than as a resale wedge.¹²

The threshold and orientation differ from costly reversibility: Abel and Eberly (1996) formalize it as a purchase price above the resale price, and Cooper and Haltiwanger (2006) estimate a mixture of convex, nonconvex, and resale-price costs. Zhang

¹²The calibration sets $\bar{\mu} = 3.05$ and $r = 3$, hence $(\mu_-, \mu_+) = (1.53, 4.57)$. The remaining annual parameters are $\beta = 0.96$, $\delta = 0.10$, and $\alpha = 0.33$; $A \in \{0.98, 1.02\}$ has transition matrix with diagonal entries 0.90. Appendix H details the calibration: the ratio is disciplined by the downside/upside semideviation ratio of annual investment growth (PNFICA, 1967–2019) rather than the raw skewness, which is fragile to the 2009 outlier, and $\mathcal{R}$ is reported across the robustness band $r \in [2, 4]$. That moment bounds r from below at about 2.3 without bounding it from above, so $r = 3$ is a conservative choice inside the identified set rather than a structural estimate of the two curvature parameters. Relative investment volatility is untargeted, since even the frictionless limit sits within one standard error of the narrow series; the level is anchored on the investment- q elasticity.

(2005) uses asymmetric quadratic curvature with $\theta^- > \theta^+$ switching at gross investment $I = 0$, which puts the high curvature on disinvestment rather than on expansion. The specification is a parsimonious setting for studying how an asymmetric real friction tilts policy graphs.

Two comparative statics follow: the first moves the friction pair (μ_-, μ_+) at fixed mean curvature; the second moves risk aversion (Figure 3). All same-capital comparisons use interior common-support domains. Within a given specification, define $\mathcal{R}(K) = C'_B(K)/C'_G(K)$ at the same K over the intersection of the regimes' 5th–95th percentile policy supports; the marginal- q analogue is $\mathcal{R}_q(K) := q_B^{K'}(K)/q_G^{K'}(K)$, the same construction at $g = q^K$. Holding $\bar{\mu} = 3.05$ fixed, the median is 1.265 under the calibrated asymmetry ($r = 3$), 0.976 under symmetric curvature ($r = 1$), and 0.752 under the reversed asymmetry ($r = 1/3$). The friction's orientation therefore determines which regime's saddle is steeper, and reversing it moves $\mathcal{R}$ from above one to below it. Under the calibrated tilt the separations narrow as capital rises: over the sampled range the consumption gap between the saddles falls by 14 percent and the marginal- q gap by 36 percent, so how well capitalized the economy is governs how far a given switch moves it.

The upper row of Figure 3 visualizes a general-equilibrium comparative static on the geometry itself: the saddle pair for $A = B, G$ is the object compared, and preferences are the parameter moved. For this exercise, utility extends from the log benchmark to CRRA, $u(c) = c^{1-\gamma}/(1-\gamma)$, with γ the coefficient of relative risk aversion. Every case keeps $r = 3$, equivalently $(\mu_-, \mu_+) = (1.53, 4.57)$ at $\bar{\mu} = 3.05$, together with the same technology, shock process, and simulated shock path. The left column reports consumption saddles $C_A(K)$; the right reports marginal $q_A^K(K) = 1 + \mathcal{C}_{K'} = 1 + \tilde{\mu}(K' - K)/K$. The top row varies the CRRA coefficient γ , changing relative risk aversion and the elasticity of intertemporal substitution (EIS), $1/\gamma$, jointly. As γ rises from 1 to 10, the median $\mathcal{R}$ falls from 1.265 to 1.047. The median $\mathcal{R}_q$ falls from 0.468 to 0.321 as both q saddles flatten. But the conditional level gap $\mathbb{E}[q^K | G] - \mathbb{E}[q^K | B]$ rises from 0.0153 to 0.0315, and the simulated unconditional $\text{sd}(q^K)$ rises from 0.00932 to 0.01723. Marginal q therefore becomes less sensitive to capital position within a regime but more volatile because regime switches traverse more widely separated saddles.

The bottom row of Figure 3 provides comparative statics with the Epstein–Zin utility. For this analysis, Epstein–Zin utility holds the EIS fixed at one while relative

risk aversion rises from 1 to 10. Strikingly, the curves barely move: the median $\mathcal{R}$ remains approximately 1.265, the median $\mathcal{R}_q$ moves only from 0.4684 to 0.4685, and $\text{sd}(q^K)$ changes only from 0.00932 to 0.00933. The strong CRRA comparative statics come mainly from the accompanying fall in the EIS, not from higher risk aversion, consistent with Tallarini (2000).

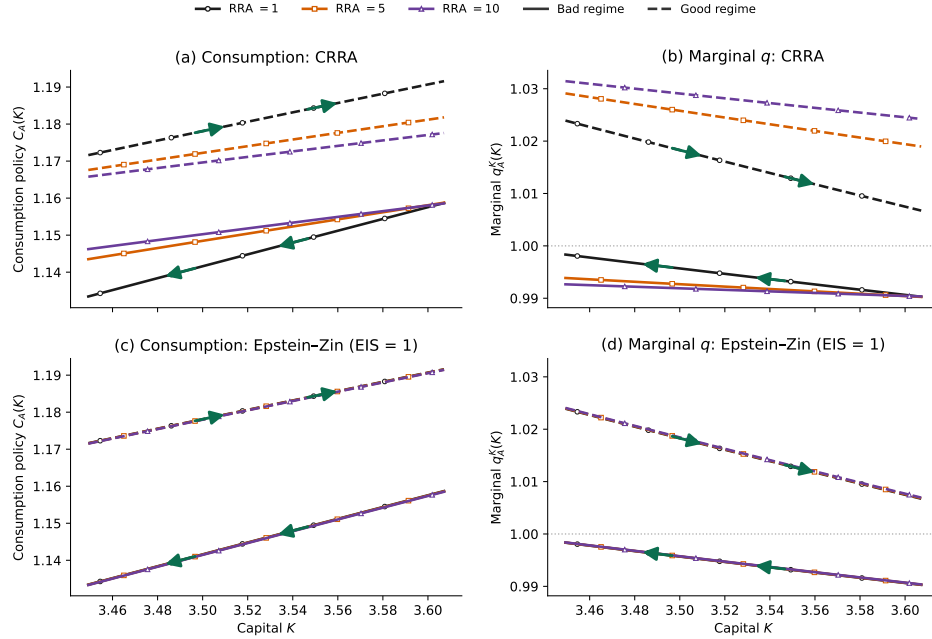


Figure 3: Consumption and marginal- q saddles across preferences

Notes: Panels (a) and (c) plot the bad- and good-regime consumption policy $C_A(K)$; panels (b) and (d) plot marginal $q_A^K(K)$. Colors and markers denote relative risk aversion; solid and dashed lines denote the bad and good regimes. The top row varies risk aversion under CRRA, the bottom under Epstein-Zin with the EIS held at one. Every curve is evaluated at the same K on the intersection of the solved supports across all regime-preference cases. Horizontal lines mark $q^K = 1$, zero net capital accumulation; arrows give the direction of capital motion, read off the sign of $q^K - 1$. Since $q_B^K < 1$ and $q_G^K > 1$ throughout, each regime's conditional steady state lies outside the plotted band and the economy is always in transit between the two. Marginal q displays the assumed adjustment wedge and is not independent calibration evidence. Same-mean symmetric and sign-reversed cases are reported in Appendix H.

Having used the asymmetric friction to isolate slope-driven state dependence, I now return to the frictionless RBC benchmark to study a distinct channel: variation in response shape generated by initial position along the saddle.

Response shape in the frictionless benchmark Although $\mathcal{R} \approx 1$ in the benchmark RBC economy, the *shape* of the response trajectory is highly state dependent. A positive productivity shock displaces the economy across saddles, and the trajectory

depends on K_0 's position relative to K_G^{cs} . Far below K_G^{cs} , the marginal product of capital is high, tilting the intertemporal tradeoff toward accumulation and producing a pronounced, late consumption hump (Barro and Sala-i-Martin, 1992); near K_G^{cs} , across-saddle reversion dominates quickly and the peak arrives earlier. This channel is orthogonal to the slope asymmetry of Proposition 5: slope differences govern *impact* magnitudes (cross-saddle dimension), position-dependent convergence speed governs response *shapes* (along-saddle dimension). The shape does *not* require a nonlinear saddle: even in the log-linearized economy, where the conditional saddle is a straight line, peak timing varies with K_0 . The mechanism is a race between the endogenous along-saddle rate λ and the exogenous across-saddle rate ρ , weighted by initial distance from steady state; concavity of $f(K)$ amplifies the effect but is not its source.

Near the non-conditional steady state, the Blanchard–Kahn decomposition (Blanchard and Kahn, 1980) on the saddle branch yields

$$\hat{c}_t = \underbrace{\mathcal{P}(\hat{k}_0)}_{\text{along saddle}} \lambda^t + \underbrace{\mathcal{Q}}_{\text{across saddle}} \rho^t, \quad (23)$$

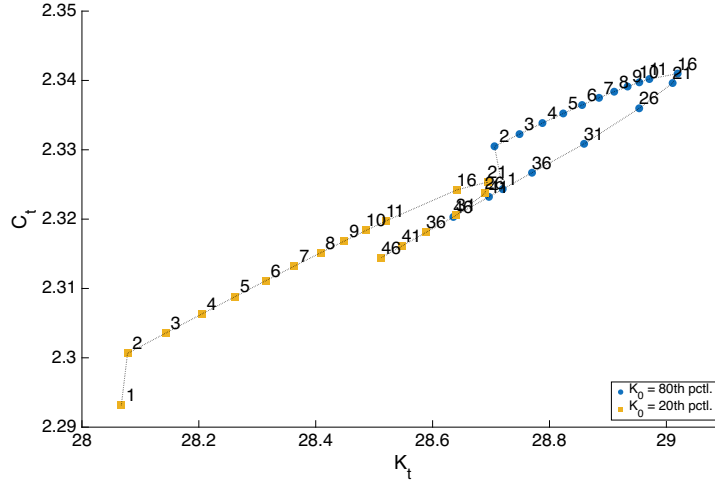
where $\hat{c}_t$ and $\hat{k}_0$ are log-deviations from steady state, $\lambda \in (0, 1)$ is the stable eigenvalue of the capital dynamics, and $\rho \in (0, 1)$ is the TFP persistence. The shock size $\varepsilon > 0$ enters only through $\mathcal{P} = \psi_k[\hat{k}_0 + \theta\varepsilon/(\lambda - \rho)]$, which depends on initial capital via the saddle slope $\psi_k > 0$ and the capital-TFP elasticity $\theta > 0$, and $\mathcal{Q} = [\psi_a - \psi_k\theta/(\lambda - \rho)]\varepsilon$, which does not.¹³ The hump arises from the tension between two decaying exponentials: the λ^t term captures along-saddle propagation, the ρ^t term direct TFP reversion, and the peak occurs where they balance. Lower initial capital reduces $\mathcal{P}(\hat{k}_0)$, weakening the slowly decaying along-saddle term, so the balance tips later, exactly the comparative static $\partial t^*/\partial \hat{k}_0 < 0$ below.

Proposition 6 (State-dependent peak timing).

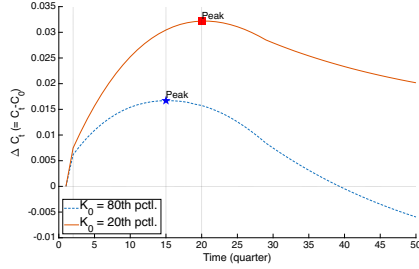
Suppose $\lambda > \rho$, $\mathcal{P}(\hat{k}_0) > 0$, and the consumption response (23) admits a peak at $t^* > 0$. Then

$$\frac{\partial t^*}{\partial \hat{k}_0} = \frac{-\psi_k}{\mathcal{P}(\hat{k}_0) \cdot \ln(\lambda/\rho)} < 0, \quad (24)$$

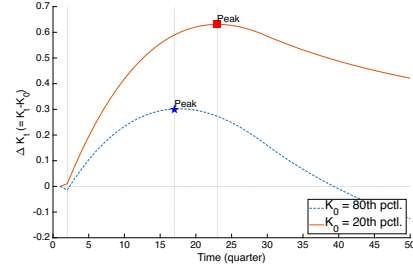
¹³This λ (at the non-conditional steady state) and the regime-specific λ_A^s of Proposition 5 (at the conditional steady state) coincide to first order in the frictionless RBC ($\lambda_B^s = \lambda_G^s = \lambda$, so $\mathcal{R} \approx 1$); frictions generate $\mathcal{R} \neq 1$. Proposition 6 then operates in the $\mathcal{R} \approx 1$ regime, its state dependence driven by initial location $\hat{k}_0$ rather than regime-varying slopes.



(a) On the saddles



(b) C : time domain



(c) K : time domain

Figure 4: Shape of the impulse responses: low K_0 vs. high K_0

Notes: Impulse responses to a positive productivity shock from K_0 at the 20th percentile and 80th percentile of the ergodic distribution. Panel (a): post-shock trajectories in (K, C) space on the conditional saddle paths; numbers indicate quarters since impact. Panels (b) and (c): time-domain responses of consumption and capital; vertical markers indicate the peak quarter. Sample: 50 quarters.

where $\hat{k}_0$ is the log-deviation of the initial capital stock from steady state.

Proof.

See Appendix B. ■

The consumption peak arrives strictly later when initial capital is farther below the steady state. The hypotheses are mild: $\lambda > \rho$ holds under standard quarterly RBC calibrations ($\lambda \approx 0.96$, $\rho \leq 0.95$), and $\mathcal{P}(\hat{k}_0) > 0$ requires only $\hat{k}_0 > -\theta\varepsilon/(\lambda - \rho)$, which the ergodic distribution satisfies with substantial margin. Proposition 6 is local by construction, and Figure 4 shows the mechanism operating globally. Comparing the 20th and 80th percentiles of the ergodic capital distribution, from low K_0 both C and

K rise longer before reversion dominates, producing a later and larger consumption peak, while from high K_0 the peak arrives earlier. The same saddle geometry, shock, and preferences apply; only the inherited capital stock differs.

4 Applications

The two applications extend the RBC benchmark of Section 2.2 in complementary directions. The heterogeneous-household economy (Section 4.1) adds idiosyncratic income risk; monotonicity lets aggregate capital index each spell of the frozen orbit, despite its true distributional state. The investment-externality economy (Section 4.2) instead adds investment-generated aggregate knowledge: the second inherited stock turns the family of frozen paths into a *surface*, whose gap from the planner’s surface identifies the distortion and optimal policy.

4.1 Heterogeneous households: dimension reduction

In a heterogeneous-household economy the natural state is the entire cross-sectional wealth distribution, an infinite-dimensional object. Classical aggregation (Gorman, 1961) reduces dimensionality by restricting preferences, but these restrictions fail under uninsurable idiosyncratic risk and borrowing constraints. Yet practitioners routinely approximate the state by a single number, aggregate capital. The conditional saddle framework explains why this works without preference restrictions by splitting the claim into two halves of very different difficulty: *within* a frozen-regime spell the distribution evolves along a one-dimensional orbit, so aggregate capital is an *exact* coordinate for it. *Across* histories, where the classical puzzle actually resides, a single capital rule works only approximately, an error this section bounds and documents directly.

I consider a continuum of ex-ante homogeneous households of unit measure, with recursive problem

$$v(a, z; \Phi, A) = \max_{c, a'} \log(c) + \beta \mathbb{E} v(a', z'; \Phi', A') \quad (25)$$

$$c + a' = a(1 + r(\Phi, A)) + w(\Phi, A)z \quad (26)$$

$$a' \geq 0 \quad (27)$$

$$\Phi' = \Gamma_{\text{endo}}(\Phi, A), \quad A' \sim \Gamma_{\text{exo}}(A'|A). \quad (28)$$

The problem differs from the representative-household economy only in (i) uninsurable idiosyncratic labor productivity (Bewley, 1986; Huggett, 1993; Aiyagari, 1994), a Markov process $z \sim \Gamma_z(z'|z)$; (ii) inclusion of the distribution Φ in the aggregate endogenous state; and (iii) the corresponding law of motion. The RCE is defined as in Krusell and Smith (1998).

The model has two exogenous stochastic processes, idiosyncratic productivity and aggregate TFP, hence two layers of conditional saddle paths, individual and aggregate. I focus on the aggregate path under stochastic TFP, following Krusell and Smith (1998).¹⁴ Consistent with Section 2.2, $K = K(\Phi)$ and $C = C(\Phi, A)$ denote aggregate capital and consumption.

The conditional K -nullcline of Definition 5 extends directly to the heterogeneous-household setting: net investment depends on Φ only through the aggregates (K, C) , so the locus remains an object in the aggregate phase plane.

Definition 6 (Conditional K -nullcline in the heterogeneous-household setting).

Fix $A \in \{B, G\}$. The conditional K -nullcline is the zero-net-investment locus in the aggregate (K, C) plane,

$$\mathcal{N}(A) := \{(K, C) \in \mathbb{R}_{++} \times \mathbb{R}_{++} : \Delta K(K, C; A) = 0\},$$

where $\Delta K(K, C; A)$ is the one-period change in aggregate capital implied by the aggregate resource constraint at (K, C) under regime A . As in the representative-agent case, $\mathcal{N}(A)$ is distinct from the realized crossings $\{(K_t, C_t) : K_{t+1} = K_t\}$ along any frozen-regime continuation $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$, which under strict K -monotonicity contain no point at all before the limit.

Krusell and Smith (1998) posits an endogenous law of motion tracking aggregate capital K rather than the full distribution Φ : $\log K' = b_0(A) + b_1(A) \log K$. This embeds two distinct assumptions: (i) aggregate capital K suffices to summarize the endogenous aggregate state, and (ii) K follows a log-linear law of motion.

Since Krusell and Smith (1998) it has been an open question whether K -based forecasting works so well ($R^2 > 0.9999$) by accident or through a deeper structural property.¹⁵ The conditional-saddle framework splits (i) into two questions of very

¹⁴In the original Krusell and Smith (1998) model exogenous individual labor supply co-moves with aggregate TFP, and the computation follows that specification. Aggregate labor L_A is constant within a regime, so output is $AK^\alpha L_A^{1-\alpha}$ and the frozen-regime analysis is the fixed-labor one with A replaced by the effective productivity $AL_A^{1-\alpha}$. I write A for that product below.

¹⁵Den Haan (2010) shows the forecasting R^2 can substantially overstate dynamic accuracy, because

different difficulty. *Within* a frozen-regime spell, K exactly indexes the distributional state on the conditional saddle, so conditioning on K loses no within-spell information; because the coordinate is strictly monotone along a single orbit, this holds path by path. The economically substantive question is the across-history one at the heart of the [Krusell and Smith \(1998\)](#) puzzle: why approximately the *same* rule $K' = f(K, A)$ works from every distributional state, not merely along one orbit. Proposition 9 bounds the across-spell discrepancy under a uniform contraction condition: the discrepancy decays with shared regime history and vanishes as regime persistence grows. Two caveats qualify the claim: the contraction condition is maintained rather than derived from primitives, and the bound is loose in the standard calibration. The near-exact accuracy is instead documented directly, by comparing the globally solved K' against the (K, A) -based prediction at spell boundaries (an average discrepancy of 2.5×10^{-4} in $\log K$). The synchronization theorem identifies the mechanism that *would* make one rule accurate across histories; the direct check confirms the accuracy is present in this calibration. The same geometry indicates where approximate aggregation should strain. When nonconvexities flatten the nullcline gap or generate multiple K -stationary points, as in lumpy-investment economies where the cross-sectional distribution carries independent forecasting power ([Khan and Thomas, 2008](#); [Bachmann et al., 2013](#)), the monotonicity hypothesis weakens. The framework then predicts degraded scalar-state accuracy rather than assuming it away.

By contrast, (ii) remains a parametric approximation (the true law of motion for K is generally non-Markovian and need not be log-linear), a matter of functional form, not coordinate choice. Within a spell richer functional forms sharpen accuracy; across spells additional moments can help, with gain bounded by the across-spell discrepancy, small in the calibration examined here.

The following theorem is the first step toward exact conditional sufficiency of K . If the economy converges along a one-dimensional equilibrium path and some aggregate variable moves strictly in one direction throughout, its current value identifies exactly where the economy is, since no two distinct distributional states share the same value.

Theorem 1 (Monotone aggregate variable as a coordinate).

Fix (Φ_0, A_0) and $A \in \{B, G\}$. Let $\{\Phi_t\}_{t \geq 0}$ be the frozen-regime continuation $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$ and let $\Phi^{cs} := \lim_{t \rightarrow \infty} \Phi_t$. Let $e : \mathcal{M}(\Phi_0, A) \rightarrow \mathbb{R}$ be an aggregate equilib-

one-step-ahead fit masks the accumulation of errors over long simulated paths; the across-spell handoff check reported below is in the spirit of his stricter multi-period standard.

rium variable. Let $t^* \leq \infty$ be the first date at which $\Phi_t = \Phi^{cs}$. Suppose that the scalar sequence $\{e(\Phi_t)\}_{t \geq 0}$ is strictly monotone for $t < t^*$ and converges to $e(\Phi^{cs})$. Then e uniquely indexes states on the conditional saddle:

$$e(\Phi) = e(\Phi') \implies \Phi = \Phi' \quad \forall \Phi, \Phi' \in \mathcal{M}(\Phi_0, A). \quad (29)$$

Proof. Define $\iota_A : \mathbb{N} \cup \{\infty\} \rightarrow \mathcal{M}(\Phi_0, A)$ by $\iota_A(t) := \Phi_t$ for $t \in \mathbb{N}$ and $\iota_A(\infty) := \Phi^{cs}$. By definition of $\mathcal{M}(\Phi_0, A)$ and $\Phi_t \rightarrow \Phi^{cs}$,

$$\mathcal{M}(\Phi_0, A) = \{\Phi_t : t \geq 0\} \cup \{\Phi^{cs}\}, \quad (30)$$

so ι_A is surjective. Take first $t^* = \infty$. Strict monotonicity of $\{e(\Phi_t)\}$ implies $e(\Phi_t) \neq e(\Phi_\tau)$ for $t \neq \tau$, hence $\Phi_t \neq \Phi_\tau$, so ι_A is injective on $\mathbb{N}$; convergence to $e(\Phi^{cs})$ gives $e(\Phi_t) \neq e(\Phi^{cs})$ for all finite t , so ι_A is injective on $\mathbb{N} \cup \{\infty\}$, hence a bijection. If instead $t^* < \infty$, the orbit is the finite set $\{\Phi_0, \dots, \Phi_{t^*}\}$, on which e takes $t^* + 1$ distinct values by strict monotonicity up to t^* , so ι_A restricted to $\{0, \dots, t^*\}$ is again a bijection. For $\Phi = \iota_A(s)$, $\Phi' = \iota_A(s')$, $e(\Phi) = e(\Phi')$ gives $e(\iota_A(s)) = e(\iota_A(s'))$; since $e \circ \iota_A$ is injective, $s = s'$ and $\Phi = \Phi'$. Thus e is injective on $\mathcal{M}(\Phi_0, A)$, and an inverse map φ_A exists on $e(\mathcal{M}(\Phi_0, A))$. $\blacksquare$

Consequently, a unique inverse map $\varphi_A : e(\mathcal{M}(\Phi_0, A)) \rightarrow \mathcal{M}(\Phi_0, A)$ satisfies $\varphi_A(e(\Phi)) = \Phi$ on $\mathcal{M}(\Phi_0, A)$, and any equilibrium object restricted to $\mathcal{M}(\Phi_0, A)$ is a single-valued function of the scalar coordinate e .

The following proposition shows that the conditional K -nullclines coincide with those of the representative-agent model.

Proposition 7 (Conditional K -nullclines of Krusell and Smith (1998)).

The conditional K -nullclines of the heterogeneous-household model are identical to those of the representative-household model and invariant over the initial distribution Φ_0 :

$$C_A^{Knull}(K) = AK^\alpha - \delta K, \text{ for } A \in \{B, G\}. \quad (31)$$

Proof.

As in Proposition 3, the stationary condition $\delta K = I$ and the aggregate resource constraint $Y = C + I$ immediately imply the form of the conditional nullclines. $\blacksquare$

As anticipated after Proposition 1, this is the distribution-invariance instance of that common root, and it is what enables dimension reduction. The boundary

governing the sign of capital accumulation, an accounting identity indifferent to *who* saves, does not move as the distribution evolves along the conditional saddle.

The invariance is specific to the neoclassical production structure with exogenous labor supply. Under separable CRRA preferences with wealth effects on labor, labor supply introduces a distributional channel into the resource constraint; the nullcline then acquires distributional dependence, blocking the analytical route to K -sufficiency (though the monotonicity diagnostic of Appendix G remains applicable).

Proposition 8 (K monotonicity and injectivity).

Fix $A \in \{B, G\}$. Under Assumptions 1 and 2, together with unique K -stationarity along the frozen- A continuation (the no-plateau condition of Section 2.2), aggregate capital converges to K_A^{cs} strictly monotonically on $\mathcal{M}(\Phi_0, A)$. Consequently, K is injective on $\mathcal{M}(\Phi_0, A)$.

Proof.

By Proposition 7, the conditional K -nullcline takes the RBC form, so strict monotonicity follows the proof of Proposition 2 verbatim with $C_A^{Knull}(K) = AK^\alpha - \delta K$. Unique K -stationarity enters locally through the stable-eigenvalue condition and globally as a pathwise hypothesis verifiable by direct inspection of the simulated frozen-regime continuation. The primitive MPC condition (Appendix E, Lemma E.1) ensures no side-switching along the continuation but does not on its own rule out non-steady-state flat steps. Since $\{K_t\}$ is strictly monotone and converges to K_A^{cs} , Theorem 1 with $e(\Phi) = K(\Phi)$ yields $K(\Phi) = K(\Phi') \implies \Phi = \Phi'$ for all $\Phi, \Phi' \in \mathcal{M}(\Phi_0, A)$. ■

Proposition 7 makes the K -nullcline *sign-determining*: the graph $C = C_A^{Knull}(K)$ partitions the (K, C) plane into regions with $\Delta K > 0$ and $\Delta K < 0$ via $\Delta K = C_A^{Knull}(K) - C$. Assumption 2 rules out discrete-time jumps across this partition, so net investment cannot reverse sign along the conditional saddle. Adding unique K -stationarity (the no-plateau condition) to rule out non-steady-state flat steps, capital dynamics are strictly monotone and each frozen-regime spell admits an exact one-dimensional representation indexed by K .

Proposition 8 makes $K : \mathcal{M}(\Phi_0, A) \rightarrow \mathbb{R}$ one-to-one, so an inverse map $\varphi_{\Phi_0, A}$ exists with $\Phi = \varphi_{\Phi_0, A}(K(\Phi))$ on $\mathcal{M}(\Phi_0, A)$.¹⁶ Any equilibrium object restricted to $\mathcal{M}(\Phi_0, A)$ then admits an exact single-valued representation as a function of (K, A) .

¹⁶This is a path-specific notion of sufficiency: for a fixed (Φ_0, A) , K uniquely indexes states on the entire conditional saddle (globally *along the transition path*).

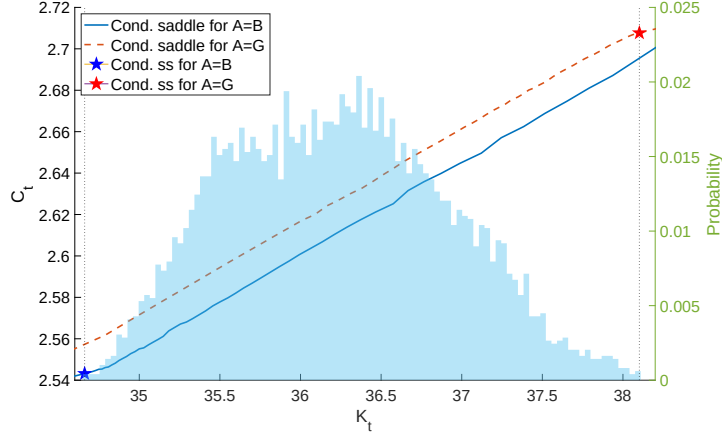


Figure 5: Conditional saddle paths in [Krusell and Smith \(1998\)](#)

Notes: The figure plots conditional saddle paths for $A = B$ (solid) and $A = G$ (dashed) implied by a canonical heterogeneous household business cycle model ([Krusell and Smith, 1998](#)). The histogram in the background plots the time-series distribution of the aggregate capital stock.

Figure 5 plots the computed conditional saddle paths in the (K, C) phase diagram. The model is solved globally using the repeated transition method, and the dynamics under each frozen aggregate state are simulated for 2,500 periods. Although the true endogenous state is Φ rather than K alone, the conditional saddle paths are strictly and monotonically ordered in K , supporting the sufficiency result.

Resolving a narrow K -window shows the ergodic band that Figure 5 displays as a line to be “dusty”. Within each spell the consecutive (K_t, C_t) points move monotonically in K , tracing an exact one-dimensional orbit, while ergodic mixing across spells produces the band observed at any given K .

From spell-wise to global sufficiency Proposition 8 is stated regime-by-regime: K uniquely indexes the conditional saddle $\mathcal{M}(\Phi_0, A)$ under a single frozen regime. The equilibrium path concatenates such one-dimensional segments, each exactly indexed by (K, A) within its spell. The handoff between spells, however, is governed by the full distributional state Φ_τ at the switch date, which (K_τ, A_τ) need not determine: different histories can arrive at the same (K_τ, A_τ) carrying different distributions. Spell-wise sufficiency is thus distinct from unconditional global sufficiency, which would require the same K to pin the same Φ regardless of path history. The following proposition bounds this across-spell discrepancy.

Observing the economy at a given K admits a “cloud” of distributions produced by different prior histories. Three results follow, with distinct jobs. Theorem 2 supplies

the mechanism: economies forget where they started, so two sharing recent regimes end up close. Proposition 9 averages that over the number of recent regimes two histories share, which persistence determines. Proposition 10 instead supplies a rate, identifying the local contraction along the saddle with the stable eigenvalue that also sets its slope.

Write $\Gamma_A := \Gamma_{\text{endo}}(\cdot, A)$. The maintained hypothesis, the substantive assumption of this subsection, is collected in the following definition.

Definition 7 (Uniform frozen-regime contraction).

The family $\{\Gamma_A\}_{A \in \{B, G\}}$ is a uniform frozen-regime contraction on $\mathcal{U}$ at rate $\lambda_c \in (0, 1)$ if $\mathcal{U} \subseteq \mathcal{X}$ is closed in the complete metric space $(\mathcal{X}, d)$, forward-invariant under every Γ_A (each Γ_A maps $\mathcal{U}$ into itself), of finite diameter $\text{diam}(\mathcal{U}) := \sup_{\Phi, \Phi' \in \mathcal{U}} d(\Phi, \Phi')$, and

$$d(\Gamma_A \Phi, \Gamma_A \Phi') \leq \lambda_c d(\Phi, \Phi') \quad \text{for all } \Phi, \Phi' \in \mathcal{U} \text{ and all } A.$$

Theorem 2 (Synchronization under frozen-regime contraction).

Let $\{\Gamma_A\}$ be a uniform frozen-regime contraction on $\mathcal{U}$ at rate λ_c (Definition 7). Then:

- (i) *(Random fixed point: the history-dependent analogue of a contraction's unique fixed point) For every regime history $\omega = (\dots, A_{t-2}, A_{t-1})$, the long-history limit (the state obtained by applying the history's maps from an ever more distant start)*

$$\Phi^*(\omega) := \lim_{T \rightarrow \infty} (\Gamma_{A_{t-1}} \circ \Gamma_{A_{t-2}} \circ \dots \circ \Gamma_{A_{t-T}})(\Phi)$$

exists, is independent of $\Phi \in \mathcal{U}$, and is the unique state in $\mathcal{U}$ consistent with ω ; any trajectory initialized in $\mathcal{U}$ at date $t - T$ satisfies $d(\Phi_t, \Phi^(\omega)) \leq \lambda_c^T \text{diam}(\mathcal{U})$.*

- (ii) *(Suffix synchronization) If two histories ω, ω' agree over their most recent T periods, then*

$$d(\Phi^*(\omega), \Phi^*(\omega')) \leq \lambda_c^T \text{diam}(\mathcal{U}).$$

Proof.

See Appendix C. ■

The limit in part (i) runs backward. Fix the recent regime sequence and start the economy ever earlier: because every map contracts, the starting state washes out and the recent history alone pins down where the economy is now. This is a random-fixed-point construction in the spirit of [Schenk-Hoppé and Schmalfuß \(2001\)](#), and part (ii) is

its consequence for two states in the same- K cloud, averaged below over the common-suffix length. For the stationary symmetric two-state chain ($\pi_{BB} = \pi_{GG} = \rho_A$), let $q := \rho_A^2 + (1 - \rho_A)^2$ denote the probability that two independent histories agreeing at a date also agree one period earlier.¹⁷

Proposition 9 (Across-spell approximation bound).

Let $\{\Gamma_A\}$ be a uniform frozen-regime contraction on $\mathcal{U}$ at rate λ_c (Definition 7), and let $\Phi, \Phi' \in \mathcal{U}$ be the long-history limits of two regime histories that agree over their most recent T periods. Then

$$d(\Phi, \Phi') \leq \lambda_c^T \text{diam}(\mathcal{U}).$$

If, moreover, the two histories are drawn independently from the stationary symmetric chain and share the current regime, then $\Pr(T = k) = (1 - q)q^k$ for $k \geq 0$ and

$$\mathbb{E}[d(\Phi, \Phi')] \leq \frac{1 - q}{1 - q\lambda_c} \text{diam}(\mathcal{U}).$$

Proof.

See Appendix C. ■

The count T begins at the previous date, because the shared current regime indexes the cloud but has not yet acted on Φ . The diameter $\text{diam}(\mathcal{U})$ is finite under the borrowing constraint $a' \geq 0$ and compact idiosyncratic support. The right-hand side involves only the primitive set $\mathcal{U}$, not the same- K cloud itself, so the proposition is a genuine bound on the cloud rather than a contraction statement about it. The bound vanishes as $\rho_A \rightarrow 1$, since long spells make recent suffixes long. It does not vanish as $\lambda_c \rightarrow 0$: a pair sharing no regime history is bounded only by $\text{diam}(\mathcal{U})$, and that happens with probability $1 - q$.

Assumed uniform contraction The hypothesis that $\Gamma_{\text{endo}}(\cdot, A)$ contracts uniformly on $\mathcal{U}$ is not innocuous: the map embeds general-equilibrium price feedback, and establishing contraction-type properties for such operators is precisely the difficulty confronted by global existence and uniqueness analyses (Proehl, 2025). Locally, Proposition 10 anchors the contraction rate to the stable eigenvalue $\lambda_A^s \in (0, 1)$; globally the assumption is maintained, and the sample-based diagnostic below provides a

¹⁷Stepping back one period, two independent histories still agree if both persist (ρ_A^2) or both switch ($(1 - \rho_A)^2$); for asymmetric chains, q is the backward agreement probability of the time-reversed chain.

lower-bound estimate, not a certification. The paper supplies no upper bound on λ_c : even $\lambda_c < 1$ is maintained rather than verified. No tension with the saddle structure arises: Γ_{endo} is the equilibrium law of motion after the forward-looking variable has been solved out, so the unstable direction of the (K, C) system is absent from the distributional dynamics.

Ergodic interpretation Theorem 2(i) licenses reading Proposition 9 on the ergodic set. Every state observed along a long simulated path is, up to a geometrically small error, the long-history limit of its own realized history, the state that history alone drives the economy toward regardless of where it started. Every pair in the same- K cloud is therefore a pair of long-history limits, inheriting the pathwise bound $d(\Phi, \Phi') \leq \lambda_c^T \text{diam}(\mathcal{U})$ with T the realized common-suffix length. This applies at every date (deep within a spell or just after a switch) because the synchronization argument runs over histories rather than single spells.

The expectation formula, by contrast, is computed under the idealized model of two independent stationary histories, and two caveats attach: within-path pairs are dependent, and conditioning on a common K is a selection. The closed form is therefore a benchmark for the typical scale of the cloud rather than a bound on its realized diameter.

Theorem 2 leaves λ_c unspecified. The next proposition pins its value near the conditional steady state: the stable eigenvalue of the linearized frozen-regime dynamics, the same object governing the local saddle slope (Proposition 5). This holds under the standard regularity of infinite-dimensional linearization: smoothness, a spectral gap, a stable manifold, and metric-norm agreement. The capital display needs one condition more, that K be C^1 and nondegenerate along the stable direction. Appendix D states all five with glosses.

Proposition 10 (Local contraction rate on the conditional saddle).

Near the conditional steady state, the frozen-regime dynamics on the conditional saddle contract distances at the geometric rate $|\lambda_A^s|$, where $\lambda_A^s \in (0, 1)$ is the stable eigenvalue of the linearized frozen-regime dynamics; in aggregate-capital units,

$$K_{t+1} - K_A^{cs} = \lambda_A^s (K_t - K_A^{cs}) + o(|K_t - K_A^{cs}|).$$

Proof.

See Appendix D. ■

The full statement on the distributional state space, its regularity conditions, and the proof are in Appendix D. Directions orthogonal to the saddle may contract at different rates, so the global rate λ_c in Theorem 2 could exceed $|\lambda_A^s|$. The proposition delivers the primitive anchor, the same eigenvalue that drives the slope-ratio diagnostic (Proposition 5) and disciplines the local contraction in Theorem 2 and Proposition 9. A certified uniform upper bound on λ_c is left to future work.

A sample-based diagnostic for λ_c The global uniform λ_c in Theorem 2 bounds contraction across a cloud of distributional states at the same K reached from different histories, which need not lie on a single conditional saddle. The diagnostic below can show the map contracts no faster than some observed rate; it cannot certify that it contracts at least that fast for every pair of states, which is what the bound needs. Since the across-spell bound uses λ_c as an *upper* bound on the per-step rate, a certified verification of Proposition 9 requires a certified upper bound on λ_c . Such a bound would come via either a Lipschitz constant for Γ_{endo} on $\mathcal{U}$ (model-specific) or a statistical confidence bound on sample estimates (standard but tedious in function spaces). This is beyond the scope of the paper and left to future work. Instead I propose a sample-based *diagnostic* that measures the one-step contraction actually exhibited by the computed map.

The diagnostic draws N pairs of distributional states at common K and regime from the simulated equilibrium, applies the frozen-regime map to both members of each pair, and reports $\hat{\lambda}_c^N := \max_i r_i$ over the one-step ratios r_i . The ratios are computed in total variation on the joint distribution over assets and employment. The map is non-expansive in that metric by construction, so $r_i \leq 1$ holds automatically and the diagnostic measures how far below one the observed rates fall. A contraction modulus depends on the metric, not only the topology, so this does not bound the λ_c of Definition 7, and no such inequality is claimed. Substituting it in the across-spell formula illustrates what the formula delivers at an observed rate rather than bounding it.

Under the Krusell and Smith (1998) calibration, quarterly regime persistence is $\rho_A = \pi_{AA} = 0.875$ (average 8-quarter spells), so $q = \rho_A^2 + (1 - \rho_A)^2 \approx 0.781$. On the simulated panel the diagnostic yields $\hat{\lambda}_c^N \approx 0.999$, with a median pairwise ratio of 0.982 over some 45,000 same- K pairs. The sample therefore exhibits almost no contraction margin, and at such a rate the benchmark formula of Proposition 9 would

leave a factor of 0.997 multiplying $\text{diam}(\mathcal{U})$, which is uninformative. The gap from the along-saddle rate is the one Proposition 10 anticipates. The deterministic benchmark saddle contracts at 0.965, but directions off the saddle contract far more slowly, and the uniform rate is set by the slowest of them. The direct quantitative check, comparing the globally solved K' at each spell boundary against the prediction from tracking only (K, A) , gives an average discrepancy of 2.5×10^{-4} in log terms (1.3% of one standard deviation of $\ln K$). This measures a mean forecast error in K' rather than a worst-case distance between distributions, so it is not directly comparable to the uniform distributional bound above; it separately documents that the across-spell error is small in this calibration.

State reduction, functional form, and block recursivity The Krusell and Smith (1998) approach is often labeled near-rational (Akerlof and Yellen, 1985), but this conflates two sources of approximation the conditional-saddle structure separates. Within an orbit, conditioning on K is exact, and only the functional form of the law of motion approximates. This within-orbit exactness connects the conditional sufficiency of K to *block recursive equilibrium* (BRE), where aggregate prices and quantities follow from a low-dimensional state (Menzio and Shi, 2010, 2011; Schaal, 2017). In directed-search environments block recursivity holds by construction; here it emerges spell-wise from the saddle geometry, with prices, quantities, and Φ recoverable from K along a given conditional saddle. Across histories, by contrast, a residual state-reduction error remains, small in this calibration, which the framework identifies rather than assumes away. This is where K -sufficiency is weaker than exact aggregation, in which (K, A) alone determines the aggregates regardless of history. Recovery from K also requires knowing which conditional saddle the economy is on, and that saddle is selected by the sequence of regimes that generated the current spell, not by (K, A) alone. Two different histories can reach the same (K, A) while carrying different distributions Φ , and hence lie on different orbits. These distributions coincide only approximately, and the agreement is closer the more the two histories share their recent regime path (Proposition 9).

Verifying monotonicity in practice Theorem 1 and Proposition 8 justify the repeated transition method (Lee, 2026a), which uses K as the conditional sufficient statistic in heterogeneous-agent models, with K indexing the equilibrium state with-

out pinning down the functional form of the law of motion. The framework also yields a practical deliverable: a cheap, computable check of the underlying monotonicity hypothesis (Assumption 2 with unique K -stationarity), one any solution method can run to certify scalar-state sufficiency, or flag its failure, before relying on it. It is verifiable two ways, both in the supplement. An *order-preservation check* (Appendix G) delivers within-spell K -monotonicity from the computed frozen-regime map where that map is scalar, and pathwise along each realized spell otherwise. A strictly increasing map with a unique stable rest point converges monotonically, and in the [Krusell and Smith \(1998\)](#) economy next-period aggregate capital is strictly increasing in current capital within each regime (rank correlation 0.9999–1.0000). Independently, primitive bounds on household MPCs and price feedback control side-switching under rational expectations, yielding exact within-spell K -sufficiency under log utility and Cobb–Douglas production. The global check is verified along a computed continuation rather than from primitives alone, with two conditions maintained explicitly, a pathwise finite-difference bound on aggregate consumption and the residual no-plateau condition (Appendix E). The logic extends to bonds, GHH labor, and multiple predetermined states, its boundary the lumpy-investment economy where nullcline invariance fails and monotonicity becomes a testable diagnostic rather than a theorem (Appendix I).

4.2 An investment externality: the conditional saddle surface

This second application extends the Section 2.2 benchmark with a second predetermined stock: investment now also builds aggregate knowledge H , which augments productivity but is not priced by firms. This combines the cumulative-investment learning of [Arrow \(1962\)](#), the nonappropriable knowledge of [Romer \(1986\)](#), and the two-stock propagation of [Cooper and Johri \(2002\)](#).¹⁸ With two predetermined stocks, frozen orbits form a surface. The planner supplies a second surface; under homogeneity both become cones and the saddle reappears in ratios.

An endogenous-productivity augmentation Relative to Section 2.2, the only addition is a second stock. Aggregate investment builds a stock of knowledge H that augments productivity, so effective productivity and the two accumulation equations

¹⁸[Benkard \(2000\)](#) documents organizational forgetting, which motivates the knowledge depreciation $\delta_H > 0$. Appendix K gives the supporting evidence, the planner derivation, the surface validation, and the replication code.

are

$$\tilde{A}_t := A_t H_t^\eta, \quad K_{t+1} = (1 - \delta)K_t + I_t, \quad H_{t+1} = (1 - \delta_H)H_t + \gamma I_t^\phi, \quad (32)$$

where $g(I) := \gamma I^\phi$ denotes the knowledge that aggregate investment generates. Because knowledge enters production only through $\tilde{A}_t$, output is $Y_t = \tilde{A}_t K_t^\alpha = A_t K_t^\alpha H_t^\eta$, the resource constraint $C_t + I_t = Y_t$ is unchanged, and Section 2's price schedules carry over with A_t replaced by $\tilde{A}_t$. The Section 2 household becomes one atomistic member of a symmetric unit continuum: firms take the knowledge stock as given, and households take its aggregate law of motion as given, so no private decision internalizes the effect of investment on H_{t+1} . Only A_t is exogenous, so freezing the regime still leaves H_t , and hence $\tilde{A}_t$, evolving. The private Euler equation is therefore the one from Section 2 evaluated at $\tilde{A}_t$, and it excludes the knowledge return:

$$\frac{1}{C_t} = \beta \mathbb{E}_t \left[\frac{1}{C_{t+1}} \{1 + r(K_{t+1}, \tilde{A}_{t+1})\} \right] = \beta \mathbb{E}_t \left[\frac{1 - \delta + \alpha A_{t+1} K_{t+1}^{\alpha-1} H_{t+1}^\eta}{C_{t+1}} \right]. \quad (33)$$

The stationary calibration satisfies $\alpha + \eta\phi < 1$; the homogeneous limit below sets $(\eta, \phi) = (1 - \alpha, 1)$. The quarterly calibration is standard, $\beta = 0.99$, $\delta = 0.025$, $\alpha = 0.36$, and $A \in \{0.99, 1.01\}$ with persistence $\pi_{AA} = 0.95$; knowledge adds $(\delta_H, \phi, \eta) = (0.10, 0.7, 0.01)$, with γ normalizing the deterministic competitive H^* to one (Appendix K). These three are illustrative rather than estimated, and the spillover is deliberately small so that the geometry rather than the size of the correction carries the argument. The steady-state credit depends on (η, ϕ) only through the product $\eta\phi$. The planner's unit-investment condition splits into a capital piece and a knowledge piece, and the credit is the second piece. Writing $m = \eta\phi$, $a = \alpha/[1 - \beta(1 - \delta)]$ and $b = \delta_H/\{\delta[1 - \beta(1 - \delta_H)]\}$, this gives $\bar{\tau} = bm/(a + bm)$. So $\bar{\tau}/(1 - \bar{\tau})$ is proportional to $\eta\phi$, while $\bar{\tau}$ itself has elasticity $1 - \bar{\tau} \approx 0.98$. The magnitudes below therefore scale with the assumed spillover to a close approximation, while the qualitative results do not.

The planner and the credit Because investment jointly produces next-period capital and knowledge, the planner carries *two* co-states, $p = V_K$ and $q = V_H$, alongside $\lambda = 1/C$:

$$\begin{aligned} \lambda_t &= \beta \mathbb{E}_t[p_{t+1}] + g'(I_t) \beta \mathbb{E}_t[q_{t+1}], & p_t &= \lambda_t \alpha \frac{Y_t}{K_t} + (1 - \delta) \beta \mathbb{E}_t[p_{t+1}], \\ q_t &= \lambda_t \eta \frac{Y_t}{H_t} + (1 - \delta_H) \beta \mathbb{E}_t[q_{t+1}]. \end{aligned} \quad (34)$$

In the standard model, substituting the investment condition $\lambda_t = \beta \mathbb{E}_t[p_{t+1}]$ into the capital envelope collapses the shadow value of capital to $p_t = \lambda_t(1 - \delta + \alpha Y_t/K_t)$. That step fails here: the investment condition carries the extra knowledge term $g'(I_t)\beta \mathbb{E}_t[q_{t+1}]$, so $\beta \mathbb{E}_t[p_{t+1}] < \lambda_t$ and the shadow value of installed capital lies strictly below that benchmark, by $(1 - \delta)g'(I_t)\beta \mathbb{E}_t[q_{t+1}]$. The knowledge return therefore enters as a *flow* on gross investment, not as a capitalized addition to the price of capital. A Pigouvian investment credit that prices the spillover,

$$\tau_t = \frac{g'(I_t) \beta \mathbb{E}_t[q_{t+1}]}{\lambda_t}, \quad (35)$$

restores efficiency. Because I_t , q_{t+1} , and λ_t all depend on the inherited stocks, the credit is itself a surface over (K, H) in each regime, the policy counterpart of the two allocation surfaces. At the calibration the deterministic planner steady state lies 3.9 percent above the competitive one in capital and 2.7 percent above it in knowledge, with a steady-state credit $\bar{\tau} = 0.0242$. Gross-investment learning makes this a flow incentive rather than a capitalized stock wedge. The credit is implemented as a gross-investment tax credit financed by a lump-sum levy. It lowers the effective price of *gross* individual investment $i_t := k_{t+1} - (1 - \delta)k_t$ to $1 - \tau_t$, with the levy $\mathcal{T}_t = \tau_t I_t$ rebated so the government budget balances each period. The credited household budget and Euler equation are

$$\begin{aligned} c_t + (1 - \tau_t)i_t &= \frac{\alpha Y_t}{K_t} k_t + w_t - \mathcal{T}_t, \\ (1 - \tau_t)\lambda_t &= \beta \mathbb{E}_t \left[\lambda_{t+1} \left\{ \frac{\alpha Y_{t+1}}{K_{t+1}} + (1 - \tau_{t+1})(1 - \delta) \right\} \right], \end{aligned} \quad (36)$$

where undepreciated capital is valued at the net-of-credit replacement price. Substituting the planner co-state conditions shows that the state-contingent credit (35) reproduces the planner's allocation exactly; fed back through this credited Euler on the computed planner path, it holds to a residual at solve accuracy (Appendix K).

The conditional saddle surface In the computed basin, each pair of initial stocks (K_0, H_0) launches a one-dimensional frozen-regime orbit $\mathcal{M}((K_0, H_0), A)$ (agents still price switches). Their closure, $\mathcal{S}_A = \overline{\bigcup_{(K_0, H_0)} \mathcal{M}((K_0, H_0), A)}$, is the consumption-policy surface over (K, H) . Figure 6(a) traces it as orbit fans. The deterministic-equation benchmark has two stable roots against one unstable, so the surface is a genuine saddle.¹⁹ The rest points (37.07, 0.983) and (38.92, 1.017) bracket the cloud's

¹⁹Appendix K reports the counts, eigenvalue moduli, and frozen-map classification: the frozen-map Jacobian is stable at B and resolution-ambiguous at G , though the dedicated G path converges.

capital range, $K \in [37.10, 38.91]$. The two stable roots separate the stocks. The fast root, 0.903, is almost purely a knowledge direction and sits at the knowledge depreciation rate $1 - \delta_H = 0.90$, so a displacement in H alone decays with a half-life of about seven quarters. The slow root, 0.965, is capital-dominated, with a half-life of about twenty. Knowledge therefore settles quickly onto the stationary locus $H = \gamma(\delta K)^\phi / \delta_H$, which is common to both regimes, and the economy then crawls along it in capital. The ergodic set is correspondingly a thin ribbon about that locus, departing from it by 0.8 percent on average against a 4.3 percent range in H , and the planner inherits the same structure. Since the mean regime spell is also twenty quarters, the economy never completes the capital adjustment before the regime switches: the rest points are attractors it approaches but does not reach.

The planner's surface and where it rests The planner faces the same two stocks, so its frozen dynamics trace a second surface over the same base, itself a saddle with two stable roots against three. The two surfaces are not usefully compared point by point, because the economies they describe do not occupy the same states: the externality moves the whole configuration rather than tilting one surface against a fixed other. What is well defined is where each surface rests. From the same frozen-regime solves, the planner's conditional rest points are (38.54, 1.010) and (40.43, 1.045), against the equilibrium's (37.07, 0.983) and (38.92, 1.017): each attractor is displaced by about 3.9 percent of capital. Since each economy clusters around its own pair, the two recurrent sets barely meet. The planner's ergodic capital range [38.59, 40.43] overlaps the equilibrium's [37.10, 38.91] only over [38.59, 38.91], and 12.7 percent of planner dates fall inside the equilibrium's range. Internalizing the spillover therefore raises consumption on average, by 0.6 percent at an essentially equal log standard deviation, though the planner saves more from any state. This compares ergodic distributions and abstracts from the transition, so it is not a welfare gain from adopting the credit.

The tax credit surface Figure 6(b) draws the optimal tax credit (35) on the same base as the allocation surfaces. It is state-dependent and countercyclical, averaging 2.45 percent in the bad regime against 2.39 percent in the good, because low investment raises the marginal knowledge return $g'(I_t)$. The same force makes it rise with capital, while diminishing returns to knowledge make it fall with H . A constant credit matches only the deterministic planner steady state and need not implement the planner

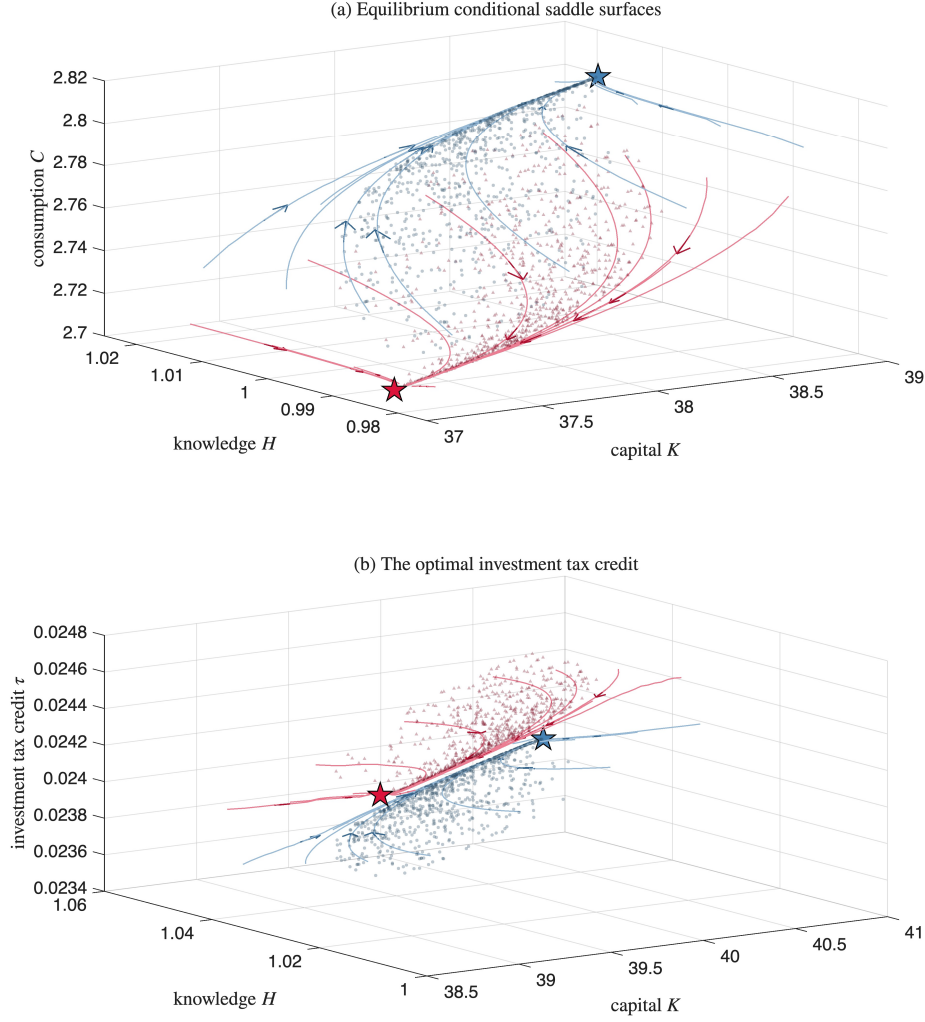


Figure 6: The conditional saddle surface and the optimal investment tax credit.

Notes: Panel (a): the good-regime (steel blue) and bad-regime (crimson) equilibrium conditional saddle surfaces, fans of frozen orbits over the sampled state region converging into the conditional rest points (stars; Appendix K), with arrows marking the direction of motion along the orbits. Panel (b): the optimal investment tax credit (35), levied on gross investment and financed by a lump-sum levy, as a surface over the same base in the good (steel blue) and bad (crimson) regimes. It is drawn in the same format: frozen planner orbits with arrows, and stars at the planner's conditional rest points. Optimal policy is itself a geometric object, countercyclical because low investment raises the marginal knowledge return.

elsewhere. The two nevertheless nearly coincide here, not because the surface is flat but because the economy travels so little across it. The surface is genuinely

tilted, with elasticities of $+0.6$ in capital and -0.5 in knowledge, yet the ergodic set spans only about four percent in each stock, so τ_t stays within $[0.0237, 0.0247]$. From a common anchor under identical shocks, $\bar{\tau} = 0.0242$ therefore tracks the planner within 0.1 percent over eight hundred quarters. *Steady-state matching is thus distinct from transition-policy implementation*, a distinction the credit surface makes explicit.

Boundary of the construction The surface exists because the composite returns satisfy $\alpha + \eta\phi < 1$, which gives two stable roots for the two predetermined stocks. At equality a unit root appears, and above the boundary only one stable direction remains, so generic two-stock convergence fails. This is a loss of saddle convergence, not sunspot indeterminacy, because this inelastic-labor economy never produces more stable roots than predetermined states. In the exact homogeneous case, $\phi = 1$ and $\eta = 1 - \alpha$, the frozen-regime system is homogeneous of degree one, the surface becomes a cone, and the ratio K/H is exactly sufficient for the normalized dynamics. This is dimension reduction by symmetry rather than by the monotonicity of Section 4.1, and it holds in all 642 realized spells of that case (Appendix K).

5 Concluding remarks

Conditional saddle paths, equilibrium trajectories under frozen exogenous regimes, bring the geometric clarity of deterministic phase diagrams to stochastic environments. Their slopes encode state dependence. Differential tilt across saddles reflects asymmetric shock responses. Their monotonicity delivers dimension reduction, since a strictly monotone aggregate variable exactly indexes the equilibrium state within each frozen-regime spell. Persistent spells, exact within-spell sufficiency, and a synchronization property under a maintained contraction condition identify the mechanism by which the conditional result would translate into near-global accuracy. The bound is loose in the calibration examined, so the near-exact accuracy is documented directly; together they give a structural account of why scalar-state methods work.

The canonical RBC model establishes the basic geometry and reveals that state-dependent response shapes arise from distance along the saddle even when slopes are nearly equal. Two applications then illustrate the framework’s reach. The heterogeneous-household model of [Krusell and Smith \(1998\)](#) demonstrates that strict monotonicity along conditional saddles delivers exact within-spell dimension reduction.

This holds locally under standard regularity, and globally on an explicit region checkable from calibrated parameters, with one residual assumption. The across-spell error is bounded, under the maintained uniform-contraction condition, by how quickly the economy forgets its regime history. A non-concave production illustration shows the no-crossing condition does not follow from feasibility alone, and a lumpy-investment experiment locates where scalar-state accuracy degrades while the one-step fit stays high. The investment-externality economy shows the geometry with a second predetermined stock: the conditional saddle is a surface, and the planner traces a second allocation surface over the same states, resting at higher capital and knowledge. Optimal policy is therefore the investment credit that implements it, itself a surface over the two stocks. At the planner’s steady state it is 2.4 percent of gross investment.

More broadly, conditional saddles offer a complementary lens on stochastic models. They provide a language for interpreting nonlinear dynamics, for assessing when scalar-state reductions are exact on the equilibrium path rather than merely accurate, and for characterizing across-path variation as a structural feature of the geometry. A promising direction is to use them to sharpen empirical restrictions on state dependence and to discipline state-contingent stabilization design.

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